

MPIM MATHEMATICS AND AI SEMINAR

Purpose. We are currently witnessing monumental impacts from the rapid development of AI on all aspects of mathematics and the world. For many, this is an exciting and frightening time. We face many urgent, but ill-formed and unsettled, questions. A few examples:

- What should the role of AI in mathematics research be? How can we protect intellectual property? What does it mean to ethically disclose AI use?
- How do we understand the value provided by human beings in the mathematical process, and how can we protect it? What is the value of a proof and of human comprehension of it, and how can we build infrastructure that rewards what we care about?
- How can mathematical pedagogy be structured to provide meaningful and robust education in the era of AI?
- What does it mean for mathematics to be accessible? How can we strive towards equity in mathematics?
- What are our obligations as citizens and human beings in a time of radical change? What mechanisms can we, individually and collectively, access to advocate for our visions of mathematics and the world?

This is equally a time of unprecedented opportunity. Many of the concerns raised by AI are not new; indeed, many of them rest upon deepset flaws in the structure of academic mathematics that we do not have the luxury to ignore. These are crucial conversations which, if we wish to control the future of our field, must happen now. We must, as a global community, actively *choose* the values we care about, and we must *work*, collaboratively, to build a more effective, inclusive, and robust mathematics.

While answering all, or any, of the questions above is beyond the scope of any single seminar or single institution, we believe it is crucial that we have a space for members of our community to develop and respectfully communicate their opinions. This is a seminar for a discussion of all topics loosely related to mathematics, AI, and the external world. Bring your ideas, your values, your anxieties, and your aspirations, and let's talk.

Time and Place. MPIM Lecture Hall, Friday 1pm-2pm.

Format. Since a goal of this seminar is to foster conversation and build community consensus, monolithic talks seem like an inefficient format for this seminar. Instead, we expect most weeks to be focused around a narrow topic or question, and to loosely follow one or both of the two following formats.

- (1) **Short exposition and Q&A.** Many in the community have specialized knowledge, or strong-held beliefs, about one of the topics proposed above and below. Speaker or speakers present on a specific topic, with ongoing questions and a substantial time for Q&A and discussion at the end. Ideally, speakers plan to talk for 30 minutes, leaving about 30 minutes for follow-up conversation.
- (2) **Guided discussion.** On some topics, it is desirable to provide a space for open discussion rather than an organized presentation. In this case, the discussion leader will prepare to oversee the discussion. For example, they might present some questions to organize the discussion, break the audience into small groups for conversation, moderate a full-room discussion, etc.

The organizers are currently soliciting volunteers for speakers or discussion leaders on all of the topics below, and are also open to suggestions for other topics. In either case, please email Cass (halleckdube @ mpim) with a short description of your topic. Since many of the topics threaten to become very broad, please focus on a narrow and specific opinion or question.

First Meeting. For the first meeting, the organizers think it would be valuable for us all to hear a broad range of inputs from the community. So if you are planning to attend, we encourage you to organize your thoughts and prepare some specific short story (a few minutes in length) about the role of AI in your life, mathematical or otherwise. Maybe you use AI in your research, and would like to talk about what form that

takes. Maybe you would like to share an emotional reaction to AI in your life or in the news. These stories are important—they help direct the rest of the seminar towards the interests and needs of the community.

Candidate topics. (In no particular order). These are of course, by no means exhaustive, and are certainly idiosyncratic to the organizers' exposure. Many important questions are being asked elsewhere, see for example the discussion [here](#). Please contact the organizers with any other topic proposal.

- Addressing AI plagiarism and intellectual property.
- Journals, publication, and disclosure of AI in mathematical research.
- Ethical and effective uses of AI for mathematical research.
- Open source vs proprietary AI: risks, benefits, user's guide.
- How to effectively use AI in mathematical pedagogy.
- Protecting education against the harmful effects of AI.
- Working for or with AI companies: personal experiences, ethics, or practical considerations.
- Collaboration between mathematics and the sciences in the era of AI.
- AI regulation and IP outside mathematics.
- Recent misalignment and breakout incidents by AI.
- Existential risk, or other discussion of a societal impact of AI.
- Mechanics for advocacy and activism, within or outside the mathematical community.

While some technical questions about LLMs, machine learning, etc., can undoubtedly arise in discussion of topics like the above, and can have real implications for AI safety, such technical questions are not intended as the focus of this seminar. We request that speakers resist the urge to give entire talks on the actual mathematics underlying AI.